

(b) $\lim_n T^n(f)$ uniformly converges to V^* . Hence V^* is weakly concave.

(c) But $V^* = TV^*$, hence V^* is strictly concave.

Q.E.D.

4 Optimal growth and its dynamics

DEFINITION 3 Inada condition on the production function: $\lim_{k \rightarrow 0} f'(k) = \infty$ and $\lim_{k \rightarrow \infty} f'(k) = 0$.

A steady state is a *constant* solution k_{ss} , such that if $k_0 = k_{ss}$, then $k_t = k_{ss}, \forall t$. Consumption is a constant sequence too. This constant sequence will have to satisfy:

$$u'(c^*) = \beta u'(c^*) f'(k_{ss}).$$

this reduces to

$$f'(k_{ss}) = 1/\beta.$$

This is the key condition for a steady state in the one-sector growth model.

In order to prove global convergence to a unique steady state, we will first state some properties of the policy function h and of steady states. Properties of $h(k)$:

1. $h(0) = 0$.
2. $h(k)$ is single-valued for all k .
3. $h(k)$ and $f(k) - h(k)$ are both strictly increasing.

$$u'(f(k) - h(k)) = \beta V^{*'}(h(k)),$$

differentiating,

$$u''(f(k) - h(k))(f'(k) - h'(k)) = \beta V^{*''}(h(k))h'(k)$$

$$h'(k) = \frac{u''f'(k)}{u'' + \beta V^{*''}} > 0.$$

Hence

$$(f'(k) - h'(k)) > 0.$$

4.

$$u'(f(k) - h(k)) = \beta f'(h(k))u'(f(h(k)) - h(h(k)))$$

implies one of the three

- $h(k) > k \iff h(k) < k_{ss}$
- $h(k) < k \iff h(k) > k_{ss}$
- $h(k) = k \iff h(k) = k_{ss}$

5. If $k < k_{ss}$, then $k < h(k) < k_{ss}$.

6. If $k > k_{ss}$, then $k > h(k) > k_{ss}$.

Before we continue with the discussion of numerical techniques, let us look at Markov Chains.

DEFINITION 4 A markov chain $\{s_t; t = 1, 2, \dots\}$ is a sequence of random variables with a state space $\{\bar{s}_i; i = 1, 2, \dots, n\}$ and with their transitional probability described by a transition matrix $\mathcal{P}$; $\Pr(s_{t+1} = \bar{s}_j | s_t = \bar{s}_i) = \mathcal{P}_{ij}$.

$\Pr(s_0)$ is initial condition

$$\Pr(s_0, s_1) = \Pr(s_0) \Pr(s_1 | s_0)$$

$$\Pr(s_0, s_1, s_2) = \Pr(s_0) \Pr(s_1 | s_0) \Pr(s_2 | s_1)$$

...

EXAMPLE 12 $\begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}$

If $\Pr(1) = \Pr(2) = 0.5$, compute $\Pr(1, 1, 1)$, $\Pr(2, 2, 2)$ and $\Pr(1, 2, 1)$.

DEFINITION 5 (Markov chain is memoryless) A markov chain $\{s_t; t = 1, 2, \dots\}$ satisfies the property that $\Pr(s_{t+1} | s_t, s_{t-1}, \dots, s_1, s_0) \equiv \frac{\Pr(s_0, s_1, \dots, s_{t-1}, s_t, s_{t+1})}{\Pr(s_0, s_1, \dots, s_{t-1}, s_t)}$ does not depend on $(s_0, s_1, \dots, s_{t-1})$. This *memoryless* property is an equivalent definition of Markov chain.

EXAMPLE 13 Suppose $s_t \in \{-1, 1\}$ satisfies $s_0 = s_1 = -1$ and $s_t = s_{t-2} \cdot s_{t-1}$. Then it is not a markov chain

EXAMPLE 14 *i.i.d. shocks* is a markov chain where $\mathcal{P}_{ij}$ is independent of i .

Value function of the sequential problem with uncertainty:

Formulation 1:

$$\begin{aligned}
V^*(x_0, s_0) = & \sup_{\{x_{t+1}\}_{t=0}^{\infty}} E \left[\sum_t \beta^t F(x_t, s_t, x_{t+1}) \right] \\
s.t. & \quad x_{t+1} \in \Gamma(x_t, s_t) \\
& \quad x_0, s_0 \text{ are given.}
\end{aligned}$$

Bellman equation

$$V^*(x_0, \bar{s}_i) = \max_{x_1 \in \Gamma(x_0, \bar{s}_i)} F(x_0, \bar{s}_j, x_1) + \beta \sum_j \mathcal{P}_{ij} V^*(x_1, \bar{s}_j), \forall x_0, \bar{s}_i.$$

Formulation 2:

$$\begin{aligned}
V^*(x_0, s_0) = & \sup_{\{x_{t+1}\}_{t=0}^{\infty}} E \left[\sum_t \beta^t r(x_t, s_t, u_t) \right] \\
s.t. & \quad x_{t+1} = g(x_t, s_t, u_t, s_{t+1}) \\
& \quad x_0, s_0 \text{ are given.}
\end{aligned}$$

Bellman equation

$$V^*(x_0, \bar{s}_i) = \max_{u_0} r(x_0, \bar{s}_i, u_0) + \beta \sum_j \mathcal{P}_{ij} V^*(g(x_0, \bar{s}_i, u_0, \bar{s}_j), \bar{s}_j), \forall x_0, \bar{s}_i.$$